

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. **Name of the Academic Unit:** Mathematics

Commented [U1]: Please enter the name in Arial 12 font

2. **Subject Name:** Optimization Lab **L-T-P:** 0-0-3
Credits: 2

Commented [U2]: Please enter the details (use Arial 12 font)

3. **Pre-requisites:** VOID

Commented [U3]: Please enter the pre-requisites (subject name and subject no. in Arial 12 font)

4. **Syllabus and reference books:**

Commented [U4]: 1. Please type the syllabus and reference books in the box provided below.

Syllabus:

Code writing in C++ for the following:

Determination of Basic Feasible Solution and selection of optimal solution,

Simplex algorithm for solving LPP, Big M method,

Dual simplex method,

Integer programming – Gomory cutting plane method ,

Application of LPP in Transportation problem; Assignment problems; Theory of Games,

Gradient Descent Methods: Steepest descent, Newton and Quasi Newton method,

Applications in statistics: log likelihood maximization problems, least squares data fitting problems

Space for entering the syllabus

Please use "Arial" font with a font size of "12" and a line spacing of 1.0 for syllabus and reference books. Please use Justified Text (Ctrl+J) for the content entered in the boxes
Reference books should be numbered as 1), 2), 3)
Please provide all bibliographic details for the reference books

Reference Books:

1)

2)

3)

5. **Lecture-wise break-up:**

Commented [U5]: Please provide lecture-wise break-up in the table given below:

Sl. No.	Topic	No. of lectures
1.	Determination of Basic Feasible Solution and selection of optimal solution	1
2.	Simplex algorithm for solving LPP	2
3.	Big M method	1
4.	Dual simplex method	1
...5	Integer programming – Gomory cutting plane method problems, least squares data fitting problems 1L	1
6	Application of LPP in Transportation problem; Assignment problems; Theory of Games	2

Please use "Arial" font with a font size of "12" and a line spacing of 1.0

Commented [U6]:

7	Gradient Descent Methods: Steepest descent, Newton and Quasi Newton methods	2
8	Applications in statistics: log likelihood maximization	2
Total number of hours		12

Commented [U7]: Please convert the template into PDF and upload it in ERP